

Tiger Lu

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APPOINTMENTS	Center for Computational Astrophysics , Flatiron Institute, New York, NY Flatiron Research Fellow	Sep 2025 –
	Yale University , New Haven, CT Graduate Researcher	Aug 2020 – May 2025
	University of California, Berkeley , Berkeley, CA Summer Undergraduate Researcher	May 2019 – Aug 2019
	Carnegie Observatories , Pasadena, CA Summer Undergraduate Researcher	May 2018 – Aug 2018

EDUCATION	Yale University , New Haven, Connecticut, USA	
	▪ Ph.D. in Astronomy with Distinction	Aug 2020 – May 2025
	• Thesis: Planets are not Points: the Profound Effect of Planetary Structure on Exoplanet System Architectures	
	• Advisers: Prof. Gregory Laughlin, Prof. Malena Rice	
	▪ M.S., M.Phil. in Astronomy	Aug 2020 – Aug 2022
	California Institute of Technology , Pasadena, California, USA	
	▪ B.S. in Astrophysics	Aug 2016 – Jun 2020
	▪ Minor in Computer Science	

PUBLICATIONS 232 citations. h -index = 9.
Italics: Student directly supervised by T. Lu

FIRST-AUTHOR

- [9] **Lu, T.**, Hernandez, D.M., Rice, M. and Su, Y. In review.
- [8] **Lu, T.** and Brown, Garrett. [A Coordinate System for Dynamical Instabilities in Hierarchical Systems in REBOUND](#). In review.
- [7] **Lu, T.**, Millholland, S.C., Rice, M., Black, B., Bardalez Gagliuffi, D.C., Balmer, W.O., Pueyo, L., Giovinazzi, M.R. and Brandt, T.D. [The Architecture of the 14 Herculis System Suggests Primordial Ejection of a Massive Planet](#). *ApJ* **1006**, 159
- [6] **Lu, T.**, Tajer, H., Hernandez, D.M., Rein, H., *Liu, Y.* and Rice, M., 2025. [Collisional Fragmentation Support in TRACE](#). *RNAAS* **9**, 5
- [5] **Lu, T.**, Li, G., Cassese, B. and Lin, D.N.C., 2025. [The Dynamical History of HIP-41378 f - Oblique Exorings Masquerading as a Puffy Planet](#). *ApJ* **980**, 39
- [4] **Lu, T.**, An, Q., Li, G., Millholland, S.C., Rice, M., Brandt, G.M. and Brandt, T.D., 2025. [Planet-Planet Scattering and ZLK Migration – the Dynamical History of HAT-P-11](#). *ApJ* **979**, 218
- [3] **Lu, T.**, Hernandez, D.M. and Rein, H., 2024. [TRACE: a Time-Reversible Algorithm for Close Encounters](#). *MNRAS* **533**, 3708
- [2] **Lu, T.**, Rein, H., Tamayo, D., Hadden, S., Mardling, R., Millholland, S.C. and Laughlin, G., 2023. [Self-Consistent Spin, Tidal and Dynamical Equations of Motion in the REBOUNDx Framework](#). *ApJ* **948**, 41
- [1] **Lu, T.** and Laughlin, G., 2022. [Tilting Uranus via Secular Spin-Orbit Resonance with Planet 9](#). *PSJ* **3**, 221

STUDENT-LEAD

- [4] *Yang, E.*, **Lu, T.**, Yahalomi, D.A. and Winn, J.N., 2026. [A Massive Hot-Jupiter Companion that Disfavors Giant Planet Formation Beyond the Water-Ice Line](#). *ApJL* **1004**, L29
- [3] *Im, H.*, **Lu, T.**, Rice, M., Tran, Q., Li, G. and Naoz, S., 2026. [Observational and Dynamical Constraints on an Unseen Outer Perturber in the GJ 436 Hot Neptune System](#). *ApJ* **1003**, 84

- [2] *Liu, Y., Lu, T. and Rice, M., 2026. Double Hot Jupiter Formation through Mirrored ZLK Migration in Binary Star Systems: The Case of WASP-94. ApJL 997, L41*
- [1] *Liu, Y., Lu, T. and Rice, M., 2025. The Formation of Double Hot Jupiter Systems Through ZLK Migration. ApJ 986, 103*

SIGNIFICANT CONTRIBUTION

- [6] Rescigno, F., Stalport, M., Anna John, A., **Lu, T.**, et al. 2026. [Understanding eccentric temperate giants: an in-depth study of the architecture and stellar obliquity of the TOI-2134 system.](#) *MNRAS* **550**, 1265
- [5] Yahalomi, D.A., **Lu, T.**, Armitage, P., Bedell, M., Casey, A.R., Price-Whelan, A. and Rice, M., 2026. [The Astrometric Resoeccentric Degeneracy: Eccentric Single Planets Mimic 2:1 Resonant Planet Pairs in Astrometry.](#) *ApJL* **999**, L9
- [4] Cassese, B., Rice, M. and **Lu, T.**, 2025. [A High-Precision, Differentiable Code for Solar System Ephemerides.](#) *PSJ* **6**, 252
- [3] An, Q., **Lu, T.**, Brandt, G.M., Brandt, T.D. and Li, G., 2025. [Significant Mutual Inclinations Between the Stellar Spin and the Orbits of Both Planets in the HAT-P-11 System.](#) *AJ* **169**, 22
- [2] Cassese, B., Vega, J., **Lu, T.**, Rice, M., Poddar, A. and Kipping, D., 2024. [squishyplanet: Modeling transits of non-spherical planets in JAX.](#) *JOSS* **9**, 100
- [1] Wahl, S., Thorngren, D., **Lu, T.** and Militzer, B., 2021. [Tidal Response and Shape of Hot Jupiters](#) *ApJ* **921**, 105

OTHER CO-AUTHOR

- [4] Naponiello, L., Poretti, E., Rice, K., et al. (inc. **Lu, T.**) 2026. [Refined parameters, formation, and dynamical stability of the wild exoplanet system K2-312=HD 80653.](#) *A&A* in press
- [3] Tajer, H., Wang, J., Childs, A., et al. (inc. **Lu, T.**) 2026. [Origins of Mercury’s Big Heart of Iron: Exploring Pathways to Form High Core Mass Fraction \(CMF\) Planets via N-body Simulations.](#) *ApJ* **998**, 46
- [2] Bardalez Gagliuffi, D.C.†, Balmer, W.O.†, Pueyo, L., et al. (inc. **Lu, T.**) 2025. [JWST Coronagraphic Images of 14 Her c: a Cold Giant Planet in a Dynamically Hot, Multi-planet System.](#) *ApJL* **988**, L18
- [1] Wang, N., Lu, L.Y., Liu, H.G., et al. (inc. **Lu, T.**) 2025. [Development of Gravity Theories in the View of TRAPPIST-1e.](#) *RAA* **25**, 015003

WHITEPAPERS

- [1] Levine, W. G., Gerbig, K., Loudon, E., **Lu, T.**, Hsieh, C., O’Connor, C., Li, R. and Dong, J., 2024. [Emerging Researchers in Exoplanetary Science \(ERES\): Lessons Learned in Conference Organization for Early-Career Researchers.](#) *Bulletin of the AAS* **56**, 2024i004

†: Joint first author

AWARDS & FELLOWSHIPS

- [Yale Dirk Brouwer Prize for Outstanding Thesis of Unusual Merit](#) 2026
- [DDA/AAS Raynor L. Duncombe Student Research Prize](#) 2025
- Flatiron Research Fellowship 2025
- Flipped Science Fair Best Poster Explanation 2024
- AAS International Travel Grant 2024
- Graduate Student Assembly Conference Travel Fellowship 2024
- National Science Foundation GRFP (*Honorable Mention*) 2022
- Joan and Arnold Seidel Griffith Observer Science Writing Contest (*Second Place*) 2022
- Alan Porter Memorial Foundation Fellow 2019
- Summer Undergraduate Research Fellow 2018

RESEARCH MENTORING

- Yurou (Nina) Liu*** Jun 2023 –
Undergraduate Researcher (*Yale*) → Ph.D. student (*Princeton*)
Papers: [Liu, Lu & Rice \(2025\)](#), [Liu, Lu & Rice \(2026\)](#)
- Haedam Im** May 2025 – Apr 2026
Dorrit Hoffleit Undergraduate Research Fellow (*MIT*)
Paper: [Im, Lu, Rice, Tran, Li & Naoz \(2026\)](#)

Eritas Yang
 Ph.D. Student (*Princeton*)
 Paper: [Yang, Lu, Yahalomi & Winn \(2026\)](#)

Oct 2025 – Jun 2026

Simone Lilavois
 Undergraduate Student (*Amherst*)

Jun 2025 –

Vighnesh Nagpal
 Ph.D. Student (*U. Chicago*)

Aug 2025 –

Haniyeh Tajer
 Ph.D. Student (*Ohio State*)

May 2025 –

*Primary Research Advisor

COLLOQUIA & SEMINARS

Italics: Upcoming

- *Astro Seminar, Carnegie EPL (Washington, DC)* Nov 2026
- *Physics & Astronomy Colloquium, SFSU (San Francisco, CA)* Oct 2026
- *Exoplanet Journal Club, University of Chicago (Chicago, IL)* Oct 2026
- *Exoplanet Seminar, Princeton University (Princeton, NJ)* Sep 2026
- *Special Seminar, UCLA (Los Angeles, CA)* Aug 2026
- *Bowler Group Meeting, UCSB (Santa Barbara, CA)* Aug 2026
- *Öberg Group Meeting, Harvard University (Cambridge, MA)* Apr 2026
- *Monday Afternoon Talk, MIT (Cambridge, MA)* Apr 2026
- *Coffee Talk, CCA (New York, NY)* Apr 2026
- *Astronomy & Planetary Science Colloquium, NAU (Flagstaff, AZ)* Feb 2026
- *Astronomy Seminar, AMNH (New York, NY)* Feb 2026
- *Powell Group Meeting, University of Chicago (Chicago, IL)* Oct 2025
- *Tea Talk, Indiana University (Bloomington, IN)* Feb 2025
- *ITC Luncheon, Harvard University (Cambridge, MA)* Jan 2025
- *Exoplanet Pizza Lunch, Harvard University (Cambridge, MA)* Jan 2025
- *Exoplanet Seminar, Princeton University (Princeton, NJ)* Nov 2024
- *Planetary Lunch Seminar, MIT (Cambridge, MA)* Nov 2024
- *Astronomy Seminar, Columbia University (New York, NY)* Oct 2024
- *Astronomy Seminar, Stony Brook University (Stony Brook, NY)* Oct 2024
- *CRA Seminar, Georgia Institute of Technology (Atlanta, GA)* Oct 2024
- *Planetary Science Seminar, Caltech (Pasadena, CA)* Oct 2024
- *Lunch Seminar, UCLA (Los Angeles, CA)* Sep 2024
- *Exoplanet Group Meeting, UCSB (Santa Barbara, CA)* Sep 2024
- *FLASH Talk, UCSC (Santa Cruz, CA)* Sep 2024
- *Star and Planet Lunch, University of Hawai'i (Honolulu, HI)* Aug 2024
- *Makino Lab Seminar, Kobe University (Kobe, Japan)* Jul 2023
- *Exoplanet Seminar, Nanjing University (Nanjing, China)* Jul 2023
- *Exoplanet Seminar, Princeton University (Princeton, NJ)* Mar 2023
- *Astronomy Seminar, Monash University (Monash, Australia)* Feb 2023

CONFERENCE TALKS

Bold: Invited Talk

- CT Exoplanet Picnic, Wesleyan College (*Wesleyan, CT*) Jul 2026
- New York Area Exoplanets Meeting III, AMNH (*New York, NY*) May 2026
- International Conference on Exoplanets and Planet Formation (*Shanghai, China*) Dec 2025
- 246th AAS Meeting Dissertation Talk (*Anchorage, AK*) Jun 2025
- Boston Area Planetary Science Meeting, MIT (*Cambridge, MA*) May 2025
- New York Area Exoplanets Meeting II, CCA (*New York, NY*) May 2025
- 56th DDA Meeting, Georgia Institute of Technology (*Atlanta, GA*) May 2025
- **ML in Planetary Dynamics Workshop**, CCA (*New York, NY*) Apr 2025
- CT Exoplanet Picnic, Wesleyan College (*Wesleyan, CT*) Jul 2024
- Emerging Researchers in Exoplanet Science IX, Cornell University (*Ithaca, NY*) Jul 2024
- **REBOUND 24 Meeting** (*Virtual*) Jul 2024
- New York Area Exoplanets Meeting I, CCA (*New York, NY*) May 2024

	▪ 55th DDA Meeting, University of Toronto (<i>Toronto, Canada</i>) ▪ 54th DDA Meeting, Michigan State University (<i>East Lansing, MI</i>) ▪ 240th AAS Meeting (<i>Pasadena, CA</i>)	May 2024 May 2023 Jun 2022
POSTER PRESENTATIONS	▪ Flatiron Scientific Exchange, CCA (<i>New York, NY</i>) ▪ 57th DDA Meeting, University of Chicago (<i>Chicago, IL</i>) ▪ Emerging Researchers in Exoplanet Science, Princeton University (<i>Princeton, NJ</i>) ▪ Exoplanets V Meeting (<i>Leiden, The Netherlands</i>) ▪ Extreme Solar Systems (<i>Christchurch, New Zealand</i>) ▪ 2022 AGU Meeting (<i>Chicago, IL</i>) ▪ Exoplanets IV Meeting (<i>Las Vegas, NV</i>) ▪ 2019 AGU Meeting (<i>San Francisco, CA</i>) ▪ 233rd AAS Meeting (<i>Seattle, Washington</i>)	Jun 2026 Jun 2026 Jun 2025 Jun 2024 Jun 2024 Dec 2022 May 2022 Dec 2019 Jan 2019
TEACHING	Warrior-Scholar Program Research Project Leader <i>One week intensive boot camp preparing military veterans for undergraduate degrees</i> Yale University Teaching Fellow ▪ ASTR 210: Stars and Their Evolution (<i>Prof. Robert Zinn</i>) ▪ ASTR 160: Frontiers and Controversies in Astrophysics (<i>Prof. Marla Geha</i>) ▪ ASTR 155: Introduction to Astronomical Observing (<i>Dr. Michael Faison</i>) Guest Lectures ▪ ASTR 5180: Galactic Dynamics (<i>Prof. Marla Geha</i>) • Lecture: <i>Astrometry in the Era of Gaia</i> ▪ ASTR 575/375: Exoplanets (<i>Prof. Malena Rice</i>) • Lecture: <i>Mean Motion Resonances</i> Dean's Tutor Tutoring Caltech students in Physics, Astronomy and Computer Science	New Haven, CT Summer 2023 New Haven, CT Spring 2023 Fall 2022, Spring 2021 Fall 2021 Fall 2026 Spring 2025 Pasadena, CA 2017-2020
OUTREACH	Yale OpenLabs Local outreach organization dedicated to educating middle school children on STEM topics. Receipt of the Seton Elm-Ivy Award. ▪ Public Relations Officer ▪ Regular Outreach Speaker • Science Cafe 2024 Speaker: <i>What if 'Cotton Candy' Planet are Actually Ring Pops?</i> • Flipped Science Fair Best Poster Explanation: <i>Probing the Shapes of Planets</i> • Science Cafe 2022 Speaker: How We Discover Exoplanets Astronomy on Tap New Haven branch of outreach program aimed at engaging local community with astronomy research ▪ Social Media Director Science in the News Local outreach organization dedicated to educating the greater community on STEM topics. ▪ Outreach Speaker • Hidden Things: Pulling Back the Veil on the Universe	2023 2022 – 2025 2022 2021
INFORMAL MENTORING	Astro Sibs Program ▪ Chris Santiago ▪ Kelyn Smith ▪ Sally Jiang	2022 – 2024 2023 – 2024 2022 – 2023
TELESCOPE PROPOSALS (CO-I)	James Webb Space Telescope – 44.7 hours ID: 11253. Fluffy or Flashy? Resolving the Mystery of the Coldest-Known Super Puff. <i>PI: Ben Cassese</i> Keck Observatory (KPF) – 0.5 nights	Cycle 5 GO Yale 2026A

	Directly Probing the Migration History of WASP-53 with Precise Orbital Constraints. <i>PI: Quang Tran</i>	
	Keck Observatory (KPF) – 2 nights	Yale 2024B
	The Search for Double Hot Jupiter Systems. <i>PI: Malena Rice</i>	
	Keck Observatory (HIRES) – 2 nights	Yale 2023A
	Probing the Exoplanet Mass Discrepancy Between the Radial Velocity and Transit Timing Methods with the Anomalously Low Density Planet Sample and Keck-HIRES (Phase 2). <i>PI: Emma Loudon</i>	
	Keck Observatory (HIRES) – 2 nights	Yale 2022B
	Probing the Exoplanet Mass Discrepancy Between the Radial Velocity and Transit Timing Methods with the Anomalously Low Density Planet Sample and Keck-HIRES (Phase 1). <i>PI: Emma Loudon</i>	
PROFESSIONAL SERVICE	Science Organizing Committee – 57th DDA Meeting	June 2026
	<i>Division of Dynamical Astronomy, University of Chicago</i>	
	Science Organizing Committee Chair – DSPS	Jan 2026
	<i>A Dynamical Synthesis of Planetary Systems, New Haven</i>	
	Organizing Committee – ERES X	July 2025
	<i>Emerging Researchers in Exoplanet Science Symposium, Princeton University</i>	
	Science Organizing Committee – ERES VIII	June 2023
	<i>Emerging Researchers in Exoplanet Science Symposium, Yale University</i>	
	Reviewer	
	<i>NASA XRP, NASA Solar System Workings Panel, A&A, AJ, ApJ, MNRAS</i>	2024 – Present
	Main Co-Organizer – Yale Exoplanets and Stars Seminar	2023 – 2025
	<i>Weekly Seminar Series</i>	
	Committee Member – ACDC	2022 – 2025
	<i>Yale University Astronomy Climate and Diversity Committee</i>	
MEDIA	Yale Daily News	Jun 2025
	New study offers a double dose of ‘hot Jupiters’	
	Popular Science	Oct 2024
	What if ‘cotton candy’ planets are actually Ring Pops?	
	AAS Author Journal Series	Nov 2022
	Tilting Uranus via Secular Spin-Orbit Resonance with Planet 9	
COMMUNITY ENGAGEMENT	New Haven Road Runners – New Haven, CT, USA	
	<i>Local USATF sanctioned running club</i>	
	▪ Public Relations Officer	Jul 2023 – Jul 2024
	▪ Racing Team Director / Elite Athlete Coordinator	Jul 2022 – Jun 2025
	NCAA Athlete – Pasadena, CA, USA	2016 – 2020
	<i>NCAA Division III Cross Country/Track & Field at Caltech</i>	
	Avery House – Pasadena, CA, USA	
	<i>Student Government</i>	
	▪ Athletics Manager	2017 – 2018
	▪ Underclassmen Counselor	2017 – 2020

[CV compiled on 2026-09-11]